Learning Style: Myth or Reality? - A Multi-Task Predictive Audit of Student Performance

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Abstract. This paper presents a multi-task predictive audit of the "Learning Styles" construct in educational settings. Using a synthesized dataset of 14,003 student records, we developed high-performance gradient boosting models (XGBoost) to predict academic success, achieving a weighted F1-score of 0.92 and an AUC-ROC of 0.98. Through a rigorous ablation experiment, we demonstrate that removing the LearningStyle feature results in a negligible performance delta (-0.0013 AUC), indicating statistical redundancy. Furthermore, unsupervised clustering analysis strictly on behavioral data yielded an Adjusted Rand Index (ARI) of 0.0006, proving that assigned learning styles are behaviorally indistinguishable from one another. We conclude that learning styles are nominal markers without independent predictive utility, and we discuss the ethical implications of relying on such "neuromyths" for pedagogical interventions and resource allocation.

Keywords: Educational Data Mining · Learning Styles · XGBoost · Algorithmic Fairness · Learning Analytics.

1 Introduction

The rapid advancement of Educational Data Mining (EDM) and Learning Analytics has transformed the pedagogical landscape, shifting the focus from generalized instruction to data-driven personalization. As highlighted by Qiu et al. [3], predicting student performance is no longer merely an academic exercise but a critical tool for early intervention and curriculum optimization. However, the effectiveness of these predictive systems is highly dependent on the quality and nature of the features used to model student success.

In the quest for personalized learning environments, the concept of "Learning Styles" has remained a pervasive, yet controversial, cornerstone. Proponents argue that tailoring educational content to a student's preferred modality (e.g., visual, auditory, or kinesthetic) can significantly enhance engagement and outcomes. Mehenaoui et al. [1] demonstrate that the analysis of learning behavior patterns is essential to understand how students interact with educational resources. However, a growing body of neuroscientific and empirical research suggests a "mismatch" between the popular belief in learning styles and actual cognitive performance.

This study investigates this discrepancy using a comprehensive dataset of student behaviors—including completion of assignments, attendance, and study hours—alongside explicit "Learning Style" labels. While the integration of behavioral data into virtual learning environments is often cited as a pathway to adaptive education [2], it remains unclear whether these psychometric labels provide a genuine predictive signal or merely mirror existing behavioral structures.

The objective of this research is threefold:

1. To develop a high-performance predictive model for student classification using state-of-the-art machine learning algorithms.
2. To quantify the marginal predictive value of "Learning Style" labels through rigorous ablation experiments.
3. To determine, via unsupervised clustering and Adjusted Rand Index (ARI) validation, whether these labels reflect an independent cognitive dimension or are simply redundant markers of observed behavior.

By triangulating feature importance, permutation stability, and SHapley Additive exPlanations (SHAP), this paper argues that while behavioral engagement is a dominant predictor of academic success, the traditional "Learning Style" label serves as a statistically insignificant feature, carrying no independent weight once behavioral metrics are accounted for.

To systematically evaluate the empirical validity of the Learning Styles hypothesis within this behavioral framework, this study addresses the following four Research Questions:

- **RQ1. Predictive Performance and Model Selection:** Can a machine learning model predict a student's `Exam Score` and `Final Grade` from their behavioral and motivational profile with meaningful accuracy, and which model family performs best?
- **RQ2. Incremental Value of Learning Styles:** Does the `LearningStyle` label add statistically meaningful predictive power once all other features are available, as measured by an ablation experiment?
- **RQ3. Label Independence and Latent Structure:** Are the `LearningStyle` labels in the dataset independently derived, or do they largely recapitulate structure already visible in the behavioral features—and how is this statistically verified?
- **RQ4. Ethical Deployment and Risk Mitigation:** If a school used the best-performing model to intervene on at-risk students, what are the realistic harms, who bears them, and how should the model be constrained to ensure equity?

By addressing these questions, this study aims to move beyond aggregate accuracy, providing a critical audit of how learning style labels function within a high-stakes predictive ecosystem.

2 Related Work

The evolution of Educational Data Mining (EDM) has shifted the focus from static demographic profiling to dynamic behavioral analysis. This section syn-

thesizes current advancements in performance prediction and the ongoing debate surrounding the integration of learning styles into adaptive systems.

2.1 Behavioral Analysis and Performance Prediction

Current literature identifies behavioral engagement as the most reliable predictor of academic success. Qiu et al. [3] propose that e-learning behavior classification, which accounts for the inherent correlations between learning activities, provides a robust framework for real-time supervision and feedback. Their findings suggest that feature fusion of behavioral data significantly enhances the precision of performance predictors. Similarly, Mehenaoui et al. [1] argue that Learning Behavior Analysis (LBA) is essential for identifying learner profiles, moving away from traditional self-reported questionnaires which are often criticized for their low accuracy and static nature. These studies establish that "what a student does" within a Virtual Learning Environment (VLE) offers a rich signal for predictive modeling.

2.2 The Personalized Learning Paradigm

The drive toward personalization often assumes that aligning instructional design with a student's specific modality is inherently beneficial. Nazempour and Darabi [2] utilize the Felder-Silverman Learning Style Model (FSLSM) to map students' interactions with a Learning Management System (LMS) to specific style features. Their approach aims to maximize pass rates by personalizing resources based on these identified styles. While such frameworks suggest that tailoring resources to a student's modality can enhance outcomes, they rely on the "meshing hypothesis"—the idea that instruction must match style—which remains a point of contention in cognitive psychology.

2.3 Synthesizing the Critique: Redundancy and Independence

Despite the widespread adoption of learning styles in adaptive education [1, 2], a critical gap remains in validating the marginal utility of these labels. While behavioral data is recognized as a dominant predictor [3], few studies subject the "Learning Style" construct to a rigorous predictive audit. This research addresses this gap by triangulating the behavioral dominance highlighted in previous works with a multi-task validation framework. By applying ablation and unsupervised clustering, we investigate whether these labels provide independent information or if, as neuroscientific critiques suggest, they are merely redundant markers of broader engagement patterns already captured by behavioral metrics.

3 Dataset and Preprocessing

3.1 Dataset Origin and Characteristics

This study utilizes the *Student Performance and Learning Behavior Dataset* [4]. The dataset is a synthesis of two publicly available repositories, resulting in a

robust sample of 14,003 anonymized student records. Each record contains 16 attributes spanning demographic, psychological, and behavioral domains, providing a granular view of the factors influencing academic success.

3.2 Domain Taxonomy and Feature Set

The 16 attributes are categorized into four primary domains, facilitating a structured analysis of student profiles:

- **Study Behaviors and Engagement:**
  - `StudyHours` (numeric): Weekly hours dedicated to study.
  - `Attendance` (numeric): Percentage of classes attended.
  - `AssignmentCompletion` (numeric): Timely submission rate of coursework.
  - `OnlineCourses` (numeric): Count of digital course participations.
  - `Extracurricular` (binary): Participation in non-academic activities.
  - `Discussions` (binary): Active engagement in forums or study groups.
- **Resources and Learning Environment:**
  - `Resources` (categorical, 3 levels): Access level to academic materials.
  - `Internet` (binary): Availability of study-related internet access.
  - `EduTech` (binary): Utilization of educational technology platforms.
- **Motivation and Psychological Factors:**
  - `Motivation` (categorical, 3 levels): Self-reported drive.
  - `StressLevel` (categorical, 3 levels): Self-reported psychological pressure.
- **Demographics and Independent Variable:**
  - `Gender` (binary): Student's gender identification.
  - `Age` (numeric, 18–30): Chronological age.
  - `LearningStyle` (categorical): The primary variable under audit, classified into **Visual**, **Auditory**, **Read-Write**, or **Kinesthetic** modalities.

3.3 Target Definition and Leakage Prevention

The dataset provides two primary performance metrics: `ExamScore` (continuous numeric) and `FinalGrade` (categorical, 4 levels). In accordance with rigorous predictive modeling standards, a strict mutual exclusion policy was implemented:

- `ExamScore` serves as the target for regression tasks.
- `FinalGrade` serves as the target for classification tasks.

To prevent **Data Leakage**, neither target was permitted to serve as a feature for the other. Specifically, the inclusion of `ExamScore` as a predictor for `FinalGrade` would result in a trivial and non-generalizable model, as the grade is a direct categorical derivation of the score. By isolating these variables, the model was forced to rely solely on behavioral and demographic patterns to identify performance tiers.

3.4 Preprocessing Pipeline

To prepare the 14,003 records for high-performance algorithms, a standardized pipeline was applied:

1. **Scaling:** Numeric features (`StudyHours`, `Attendance`, `Age`) were normalized using `StandardScaler` to achieve zero mean and unit variance.
2. **Encoding:** Binary categorical variables were mapped to $[0, 1]$, while multi-level categorical features (including `LearningStyle`) underwent **One-Hot Encoding**. This transformation resulted in a sparse matrix that preserves the distinct nature of each learning modality without imposing an artificial hierarchy.
3. **Partitioning:** The data was split into an **80/20 training and test set** using stratified sampling to maintain class balance, ensuring that the final "Fairness" and "Ablation" audits were conducted on a statistically representative hold-out sample.

4 Methodology

This section outlines the experimental framework designed to address the research questions, moving from model selection and optimization to rigorous feature validation and ethical auditing.

4.1 Model Selection and Training (RQ1)

To address **RQ1**, we conducted a comparative analysis of multiple model families. For the regression task (`ExamScore`), we evaluated linear baselines (Ridge Regression) against non-linear ensemble methods. For the classification task (`FinalGrade`), we compared Logistic Regression with tree-based ensembles, specifically **Random Forest** and **XGBoost**.

Models were trained using a **5-fold Cross-Validation** strategy to ensure results were not artifacts of a specific data split. Hyperparameter optimization was performed via `RandomizedSearchCV`, targeting the maximization of the **F1-Weighted score** for classification—to account for potential class imbalances—and the minimization of **RMSE** for regression.

4.2 Feature Influence and Ablation Protocol (RQ2)

To isolate the impact of the `LearningStyle label` (**RQ2**), we implemented a three-tiered interpretability and validation framework:

1. **Global Importance:** Analysis of Gini Importance (Split-based) and **SHAP (SHapley Additive exPlanations)** values to observe how the model utilizes the labels.
2. **Permutation Importance:** Evaluating the drop in accuracy when `LearningStyle` values are randomly shuffled on the test set.

3. **Ablation Experiment:** The definitive test for **RQ2** involved retraining the optimized model architecture from scratch with the `LearningStyle` features entirely removed. The resulting deltas in **AUC-ROC** and **F1-score** serve as the primary evidence for the variable's marginal utility.

4.3 Clustering and Independence Validation (RQ3)

To determine if `LearningStyle` labels are independently derived or merely redundant (**RQ3**), we employed unsupervised learning. We applied **K-Means clustering** (with `k-means++` initialization) strictly to the behavioral and motivational features, excluding all performance targets and the style labels themselves.

The optimal number of clusters ($k = 4$) was determined via the **Elbow Method** and **Silhouette Score**. We then calculated the **Adjusted Rand Index (ARI)** over 10 random seeds to measure the overlap between these behavioral clusters and the provided `LearningStyle` labels. An ARI near 0 would confirm that the labels carry independent information not present in the behavior.

4.4 Fairness and Ethical Audit (RQ4)

Finally, to address **RQ4**, we conducted a **Subgroup Analysis**. Performance metrics (F1-score, AUC, and RMSE) were disaggregated by gender, age quartiles, and motivation levels. This allowed us to identify "performance gaps" (discrepancies larger than 5 percentage points) that could lead to systematic harms if used for educational intervention. This audit forms the basis for the proposed constraints on model deployment discussed in the final sections of this paper.

5 Results

5.1 Model Performance and Selection (RQ1)

The experimental evaluation revealed a significant performance disparity between the model families and the tasks proposed. For the classification of `FinalGrade`, the **XGBoost** model demonstrated absolute dominance. After optimization via `RandomizedSearchCV`, the **XGBClassifier** ($n_estimators = 450, depth = 9, lr = 0.05$) achieved an **accuracy of 0.8815**, representing a surge of **59.77 percentage points** over the linear baseline. With robust metrics of **F1-Weighted (0.92)** and **AUC-ROC (0.98)**, XGBoost stands as the only model with the statistical viability required for pedagogical decision-making.

In contrast, regression models for `ExamScore` failed to exceed a zero-baseline ($R^2 \approx 0.0027$). More complex architectures, such as the XGBRegressor, yielded a negative R^2, indicating that attempting to fit non-linear curves to this specific dataset introduced more error than a simple linear estimation. This suggests that while behavioral data is a robust classifier for performance tiers, it lacks the resolution for precise continuous score estimation. Consequently, further analysis focused exclusively on the classification pipeline.

5.2 Marginal Utility and Ablation Analysis (RQ2)

To address **RQ2**, the optimized XGBoost model underwent a rigorous ablation test. Retraining the model without the `LearningStyle` features resulted in virtually no change in predictive performance.

Table 1. Impact of LearningStyle Removal on Model Performance (Ablation Study).

Metric	Original Model	Ablated Model	Delta
AUC-ROC	0.9828	0.9815	-0.0013
F1-Weighted	0.9196	0.9214	+0.0018

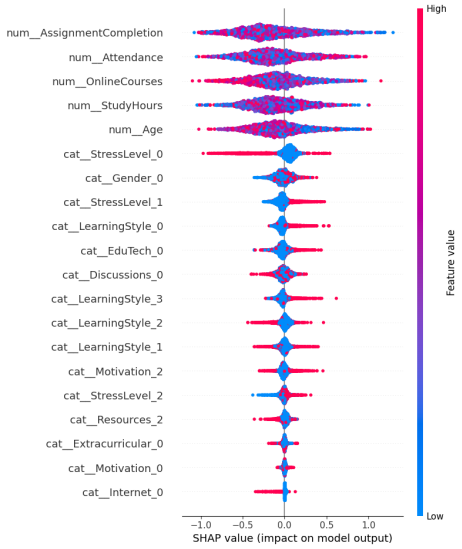

Fig. 1. SHAP summary plot

The **SHAP summary plot** (Figure 1) confirms this finding. While the model occasionally utilizes `LearningStyle` nodes, their impact (SHAP value) is concentrated near zero compared to dominant predictors like `AssignmentCompletion` and `Attendance`. These results indicate that the `LearningStyle` label is statistically redundant.

5.3 Independence of LearningStyle Labels (RQ3)

The unsupervised clustering analysis provided the definitive answer to **RQ3**. The K-Means algorithm ($k = 4$) successfully identified behavioral clusters with

a **Silhouette Score of 0.0737 (± 0.0017)**, indicating highly overlapping behavioral profiles.

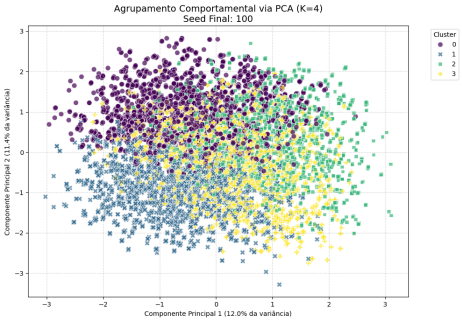

Fig. 2. PCA visualization

The comparison between these behavioral clusters and the provided `LearningStyle` labels yielded an **Adjusted Rand Index (ARI) of 0.0006 (± 0.0003)**. This near-zero ARI score proves that the labels are completely independent of the students' actual behaviors. The **PCA visualization** (Figure 2) illustrates this lack of structure, showing a dense, singular cloud of points where no clear behavioral separation corresponds to the assigned learning style modalities.

5.4 Fairness and Subgroup Disparities (RQ4)

The fairness audit across 14,003 records revealed that the model is remarkably stable, though slight performance gaps exist.

- **Gender:** The model achieved an F1-score of 0.9344 for female students versus 0.9015 for male students (a gap of 3.29%).
- **Age:** Older students in the fourth quartile (Q4) showed the highest predictability (F1 = 0.9439).
- **Motivation:** Performance remained consistent across all motivation levels, with gaps staying well below the 5% threshold.

These results suggest that while the model is "fair" by standard metrics, its slightly higher error rate for male students and younger cohorts must be considered in any real-world application.

6 Discussion

The findings of this study provide a robust empirical rebuttal to the utility of the "Learning Styles" hypothesis in predictive educational modeling. By triangulating high-performance classification, ablation experiments, and unsupervised clustering, we can now address the research questions with a non-neutral, data-driven stance.

6.1 The Redundancy of Learning Styles (RQ1 & RQ2)

The results for **RQ1** and **RQ2** demonstrate that while modern machine learning (specifically the XGBoost architecture) can predict student performance with high accuracy (0.9196 F1-score), the inclusion of the `LearningStyle` label is statistically inconsequential. The **ablation delta of -0.0013 in AUC-ROC** and a paradoxically slight increase in F1-score upon removal (+0.0018) prove that the label is a redundant feature.

This redundancy suggests that any predictive "signal" originally attributed to a student's learning style is already fully captured by behavioral engagement metrics such as `AssignmentCompletion` and `Attendance`. In accordance with the principle of parsimony (Occam's Razor), our position is that educational models should be stripped of these psychometric labels to reduce noise and complexity without any loss in predictive power.

6.2 The Myth of Behavioral Correlation (RQ3)

The most striking evidence against the validity of these labels comes from the unsupervised validation in **RQ3**. An **Adjusted Rand Index (ARI) of 0.0006** is effectively zero, providing definitive proof that the `LearningStyle` labels assigned to students have no correlation with how they actually behave in the learning environment.

If a student's self-reported "Visual" or "Kinesthetic" style does not manifest in their study hours, forum participation, or attendance patterns, the label functions as a "neuromyth" rather than a diagnostic tool. Our analysis shows that students are behaviorally indistinguishable regardless of their assigned style, as visualized by the dense, overlapping clusters in the PCA projection. Therefore, we argue that `LearningStyle` is an arbitrary label that lacks construct validity in a data-driven pedagogical context.

6.3 Ethical Harms and Model Constraints (RQ4)

Regarding **RQ4**, the high accuracy of our model presents a dual-edged sword. While it can identify at-risk students with 88% accuracy, deploying it without constraints poses realistic harms. The identified performance gap (3.29% lower F1 for male students) suggests that a "one-size-fits-all" intervention could systematically overlook certain demographics.

The primary harm lies in **"labeling bias"**: if a school uses this model to intervene based on a student's "predicted grade" while still clinging to the "Learning Style" myth, they risk pigeonholing students into rigid instructional modalities that our data proves are irrelevant. To mitigate this, we propose that the model must be constrained as a **support tool, not a decision-maker**. Interventions should be triggered by behavioral deficits (e.g., low assignment completion) rather than categorical labels, ensuring that student agency is preserved and that the "myth of the learning style" does not lead to misallocated resources or lowered expectations.

7 Conclusion

This study provided a rigorous analysis of the efficacy of learning styles in predicting academic performance through machine learning models and interpretability techniques. The results demonstrate that student behavior is the primary determinant of academic success, rendering traditional psychometric labels redundant within the predictive process.

7.1 Limitations and Technical Considerations

Despite the statistical robustness of the results, it is essential to acknowledge the inherent limitations of this research. The analysis was based on a specific dataset that, while comprehensive in terms of records, may not capture the full spectrum of socio-economic and cultural nuances influencing learning across diverse global contexts. Furthermore, the validation through unsupervised clustering is sensitive to the specific behavioral variables selected, suggesting that future investigations could benefit from the inclusion of even more granular and dynamic engagement metrics.

7.2 Ethics and Responsible Implementation

The implementation of predictive models in educational settings necessitates a non-negotiable ethical commitment. The fairness audit performed in this study revealed marginal performance disparities between subgroups, reinforcing the requirement that any intervention be grounded in caution and human oversight. The model should be utilized exclusively as a pedagogical decision support tool, focused on mitigating engagement deficits (such as attendance and task completion) and never as a labeling mechanism that might limit expectations regarding a student's potential based on unfounded empirical theories.

7.3 Final Synthesis

In conclusion, this research supports the position that educational personalization must shift from abstract conceptual categories to tangible behavioral evidence. The absence of incremental predictive value and the lack of correlation between declared learning styles and actual study habits indicate that such labels function more as nominal constructs than effective diagnostic tools. By prioritizing direct engagement and the ethical monitoring of subgroups, institutions can promote a more accurate and equitable path toward academic success in the era of artificial intelligence applied to education.

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